Waste Management 154 (2022) 105–112

Contents lists available at ScienceDirect

Waste Management

journal homepage: www.elsevier.com/locate/wasman

Improved valorization of sewage sludge in the circular economy by anaerobic digestion: Impact of an innovative pretreatment technology

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ARTICLE INFO

Keywords:
Enhanced anaerobic digestion
Sewage sludge pretreatment
Advanced wet-oxidation and steam explosion
Thermophilic anaerobic digestion
Optimization of pretreatment

ABSTRACT

Anaerobic digestion (AD) of sewage sludge shows low carbon conversion efficiency (CCE) due to the poor biodegradability of sewage sludge. The lack of digestibility is specifically linked to the waste-activated sludge (WAS) making up the majority of sewage sludge along with a smaller portion of primary sludge, depending on the wastewater treatment plant configuration. In this study, we examine the Advanced Wet Oxidation & Steam Explosion process (AWOEx) for improving the CCE of digested sewage sludge (DSS) by thermophilic AD. The effect of the pretreatment temperature in the range between 160 and 185 °C at a fixed residence time of 20 min with and without oxygen added at a dosage of 5 % of the organics present was tested. Methane yield improved by 97.92 % to 183.91 ± 4.93 mL/g vS over the untreated DSS (control), whose methane yield was 92.92 ± 9.07 mL/g vS We have demonstrated for the first time that 84 % of the organics in sewage sludge can successfully be transformed into biogas following AWOEx pretreatment, which can contribute significantly to the circular economy instead of greenhouse gas emissions from landfilling.

1. 1.Introduction

More than 13.84 million US dry tons (12.56 million metric dry tons) of sewage is produced annually in the US from publicly owned treatment works or wastewater treatment facilities (WWTF) (Seiple et al., 2017; Venkatesan et al., 2015). Sewage sludge, also referred as biosolids, by the U.S. Environmental protection Agency (U.S. EPA) is a byproduct from the WWTFs that needs disposal. Also, it should be noted that sludge management may demand a significant share of the WWTF's overall energy requirement (Capodaglio & Olsson, 2020). Though several technologies are being developed for valorizing sewage sludge, implementing these solutions at small-scale communities is still a challenge due to unfavorable economics. However, of late, under various parameter variables (season, temperature, and ambient humidity), a completely solar-powered greenhouse facility for sewage sludge drying was examined, and potential improvements for its efficiency were assessed (Boguniewicz-Zablocka et al., 2021). Due to its high organic content, digested sewage sludge (DSS) can be used to produce bioenergy and biofuels. Around 50 % of the US's WWTFs, corresponding to 7,500 total WWTFs, convert their sewage sludge to biogas, a mixture of methane (ca. 60 %) and carbon dioxide (ca. 40 %) using anaerobic digestion (AD). Analyses have shown that the digestate produced after AD contains a large portion of organics, which needs further processing before conversion to valuable bioenergy or bio-products (Li et al., 2017; Nordell et al., 2021; Wu et al., 2020). Several thermal pretreatment technologies have been developed for enhancing AD of DSS (Cambi THP, Veolia Exelys) (Abu-Orf & Goss, 2012; Bora et al., 2020). However, these methods are tailored for upfront pretreatment of the sewage sludge and not on specific treatment of the recalcitrant fraction of sewage sludge. Also, these technologies use far less severe operational conditions (temperatures of ca. 160 °C), and the biogas yields are generally only improved by ca. 30 %. (Nguyen et al., 2021; Seiple et al., 2020; Wang et al., 2021; Yan et al., 2022). Methane yields from the conventional AD of sewage sludge have been reported to range between 0.19 and 0.24

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https://doi.org/10.1016/j.wasman.2022.09.035
Received 26 July 2022; Received in revised form 17 September 2022; Accepted 30 September 2022
Available online 10 October 2022
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$Nm^3/kg\ VS_{in}$ (Di Capua et al., 2020) and the residual organics are generally concentrated and disposed of in landfills, on agricultural land and parks, or by incineration (Nguyen et al., 2021). However, some thermochemical processes like pyrolysis can recover nutrients from digestate by converting the residual organics into energy rich gas streams, promoting circular economy. Though this is more sustainable than incineration, it require further processing to close the carbon loop (Callegari & Capodaglio, 2018).

To overcome these shortcomings, we implemented Advanced Wet Oxidation and Steam Explosion (AWOEx) (a thermochemical procedure integrating wet oxidation and steam explosion) (Ahring & Munck, 2013; Miklos et al., 2018; Svensson et al., 2018; Wu et al., 2020) followed by a secondary AD (AD2).

In the current pretreatment process i.e., AWOEx, biomass materials are subjected to oxidizing agents such as air, hydrogen peroxide, or oxygen under high temperatures (over 160°Celsius) and high pressures (over 10 bars) for 15 to 45 min (Ahring & Munck, 2013; Biswas et al., 2015; Biswas et al., 2020; M'Arimi et al., 2020). The AWOEx process significantly enhances the solubilization of lignin and hemicellulose in lignocellulosic biomass including the crystalline cellulose that is retained in the solid fraction (Biswas et al., 2020; Khan & Ahring, 2020). As a result of the AWOEx pretreatment, microbial cell wall is disrupted and complex structures are broken down. As a result the digestibility of the sludge improves resulting in better degradation and disruption of both extracellular and intracellular organic matter during the hydrolysis step of AD. Thus after pretreatment, hydrolytic enzymes of the microbial population will be more effective leading to an increased methane yield (Pilli et al., 2020; Sudalyandi & Jeyakumar, 2022; Yadav et al., 2022). Additionally, acetic acid formed via de-esterification of the acetyl groups, affects the hemicellulosic dissolution into the aqueous solution (Biswas et al., 2020; Biswas et al., 2014). The AWOEx pretreatment also has several advantages besides its effectiveness, such as requiring no chemicals to be recovered, along with a lower formation of inhibitory by-products. It is possible to perform AWOEx pretreatment at dry matter concentrations up to 45 %, lowering the cost of the pretreatment (Biswas et al., 2020). Despite extensive testing of AWOEx on various lignocellulosic biomass materials, the process has not been previously studied with sewage sludge as the raw material. In the current study, DSS that has already been digested using conventional single-staged AD before dewatering and the solids have been pretreated using the AWOEx pretreatment method. The pretreated sludge is then fed into a second AD step, resulting in a two-staged AD. This process provides a way to make the recalcitrant fraction of the sludge available for digestion.

Many researchers have investigated recycling and reuse of sludge from WWTPs as a viable sustainable choice since there is an increased focus on mitigating the potential negative effect of land applying sewage sludge as well as the need for reducing landfilling of wet organic waste materials (Gupta et al., 2021; Smol et al., 2015). In the circular economy perspective, the prime focus is on re-use of the waste as a raw material, which not only avoids the costs associated with waste disposal but instead valorizes the waste as the products and byproducts trickle into the economy. Therefore, sludge reuse with a focus on energy recovery is a prominent waste management option and a cornerstone of circular economy for WWTP (Eliche-Quesada et al., 2011). In this study, we determined the effect of AWOEx pretreatment on DSS and optimized the pretreatment temperature for sustained high methane yield by thermophilic AD. To understand the impact of AWOEx on the specific components of the DSS, we performed composition analysis of the DSS out of the secondary AD (AD2).

2. Materials and methods

2.1. Advanced wet oxidation & steam explosion pretreatment

DSS was collected from the Walla Walla WWTP in Eastern Washington and then dewatered using a filter press before pretreatment. All samples of DSS before and after pretreatment were stored at 4 °C before use.

Table 1
AWOEx conditions for dewatered anaerobic DSS.

Run	Temperature (°C)	Residence time (min)	O_2 dosage (%)
1	160	20	5
2	165	20	5
3	170	20	5
4	175	20	5
5	180	20	5
6	185	20	5
7	160	20	0
8	165	20	0
9	170	20	0
10	175	20	0
11	180	20	0
12	185	20	0
13	Dewatered AD Sludge (Without Pretreatment)		

The AWOEx pretreatment of DSS was carried out in a 100 L stainless steel pressure tank. In each run, 10 kg of dewatered DSS with initial total solids conc of ca. 11 wt% was fed into the reactor at a constant mixing speed of 50 rpm. The target temperature was achieved by steam addition into the pretreatment reactor while the temperature was kept constant using an external jacketed oil heater. Oxygen (O_2) was purged into the reactor based on the amount of vS of DSS while increasing the mixing speed to 85 rpm. At the end of residence time, the material was flashed out of the tank into a 250 L flash tank. After cooling, the sample was collected and stored at 4 °C for future experiments and analysis.

2.2. Design of experiments and statistical analysis

The Advanced Wet Oxidation & Steam Explosion (AWOEx) pretreatment of DSS was performed at temperatures ranging from 160 to185°C, without and with oxygen (5 % of the VS) at a residence time of 20 min. Using the design of experiments (DOE) module (JMP Pro 15.2.0 - SAS Institute Inc. Cary, NC, USA), 12 experiment conditions at temperatures from 160 °C to 185 °C with 5 °C interval, without and with oxygen added (5 % of the VS) and a control without any pretreatment (Table 1) are defined based on the full factorial design for combined testing of the following categorical factors: temperature (6 level – 160 °C, 165 °C, 170 °C, 175 °C, 180 °C, 185 °C) and effect of oxygen addition (2 level – without oxygen and with oxygen at 5 % of VS) by recording the biogas produced for each condition.

2.3. Anaerobic digestion of pretreated DSS

Anaerobic digestion (AD) of DSS was conducted in a series of 1 L bioreactors (in duplicates) operated in semi-continuous mode with a working volume of 400 mL and with a water displacement setup to record the biogas volume produced from each reactor. The initial inoculum used was the raw sludge procured from the Walla Walla WWTP and digested sludge from the same plant mixed at a ratio of 1:1. The vS loading was always maintained at 70 wt% of the TS. The hydraulic retention time (HRT) was kept constant at 16 days throughout the experiments, and all bioreactors were fed every 48 h with a recording of the biogas produced. One bioreactor was set as control and was fed with DSS without pretreatment. All the experiments were run at thermophilic conditions (55 °C) for total of three retention cycles (48 days). Upon stable experimental data, particularly volume and composition of biogas, AD experiments were considered complete at the end of the stable retention cycle. The methane yield data collected from the experiments was analyzed using two-way analysis of variance (ANOVA) test (JMP Pro 15.2.0, SAS Institute Inc. Cary, NC, USA) to determine the effect of pretreatment conditions and its significance on improving the

Table 2
Characteristics of dewatered anaerobic DSS.

Parameters (wt %)	Values
Moisture (%)	86.4
Total solids (%)	13.6
Volatile solids (%)	5.78
Ash (%)	2.92
Cellulose (%)Hemicellulose	2.3
(%)Lignin	2.9
(%)	0.9
Crude Fat (%)	1.69
Protein (%)	3.01
Calories (Kcal/100 g)	15

methane yield. All the presented data and calculations are based on the experimental data of the stable retention cycle for all the 12 independent conditions in duplicates (repetitions). When applicable, the averages and standard deviation (±S.D) were calculated and reported.

2.4. Composition analysis

The moisture contents (MC), total solids (TS), volatile solids (VS), carbohydrates (cellulose and hemicellulose), lignin, and lipid content of all samples before and after AWOEx of DSS, DSS and AD were measured following the National Renewable Energy Laboratory (NREL) methods as previously described studies (Biswas et al., 2020; Hames et al., 2008; Van Wychen et al., 2016). Protein contents of all samples before and after AWOEx of DSS and DSS were measured by the Lowry method (Amin et al., 2021; González López et al., 2010). NREL standard method was used for determining total lignin (acid-soluble and acid-insoluble) and total cellulose and hemicellulose (Sluiter et al., 2008). Acid soluble lignin was determined using a UV spectrophotometer at 205 nm (Korpinen et al., 2014), whereas total carbohydrates (sum of glucose, xylose, and arabinose) were determined after two-step acid hydrolysis (Jenway 6405 UV/Visible, NJ, USA) at a wavelength of 320 nm (Biswas et al., 2020; Khan & Ahring, 2020). Monomers were analyzed by using an UltiMate® 3000 HPLC system (Dionex, Sunnyvale, CA) with the Aminex® 87H Column 250 × 4.6 mm (Bio-Rad, Hercules, CA) with a Shodex RI-101 refractive index detector at 60 °C using 4 mM sulfuric acid as an eluent at a flow rate of 0.6 mL/min. All measurements of lignin and carbohydrates analysis were done in duplicates. The lipid extraction was performed following chloroform–methanol extraction method proposed by Bligh and Dyer (Bligh & Dyer, 1959). 1 g of dried sludge was mixed with 20 mL of mixed liquor (chloroform: methanol = 2:1, v/v) and then subjected to 60 °C for 4 h. Thereafter, the mixture was placed in a centrifuge for 15 min at 3000 rpm and the solvent phase was transferred to a glass vial. After evaporating the solvent at 60° C, the vial was weighed. The difference in weight was the total lipid content.

2.5. Volatile fatty acids analysis

The concentration of volatile fatty acids was determined in the influent and effluent samples for each bioreactor at the start and end of every retention cycle using high performance liquid chromatography (HPLC) as previously described (Khan & Ahring, 2020). 2 mL of each sample was taken and centrifuged at 10, 000 RPM for 10 min. The supernatant was separated with pipette, diluted 2 times with MiliQ water and then filtered through 0.2 μm syringe filter. The samples were analyzed by UltiMate® 3000 HPLC system (Dionex, Sunnyvale, CA) with an Aminex® 87H column 250 × 4.6 mm (Bio-Rad, Hercules, CA) with a Shodex RI-101 refractive index detector using 4 mM sulfuric acid as eluent.

2.6. Biogas analysis

The composition of methane and carbon dioxide from all bioreactors

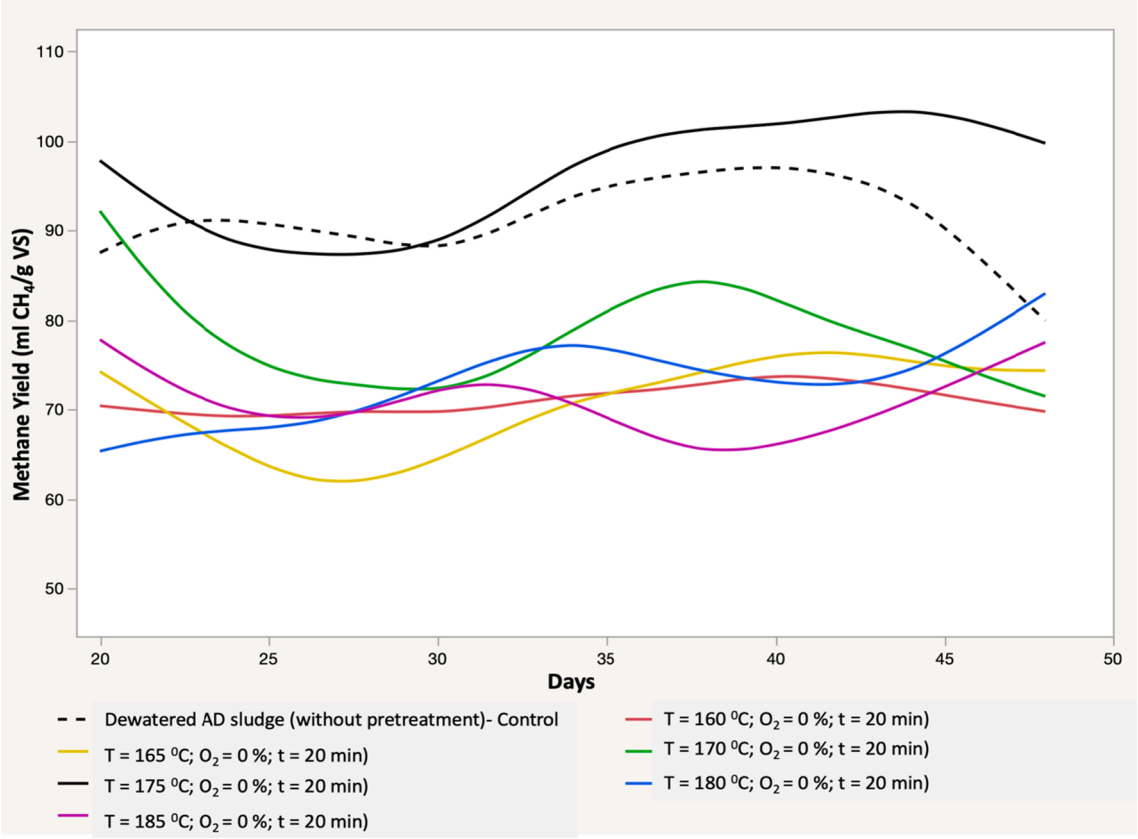


Fig. 1. Average methane yield of bioreactors fed with AWOEx pretreated DSS material pretreated without oxygen.

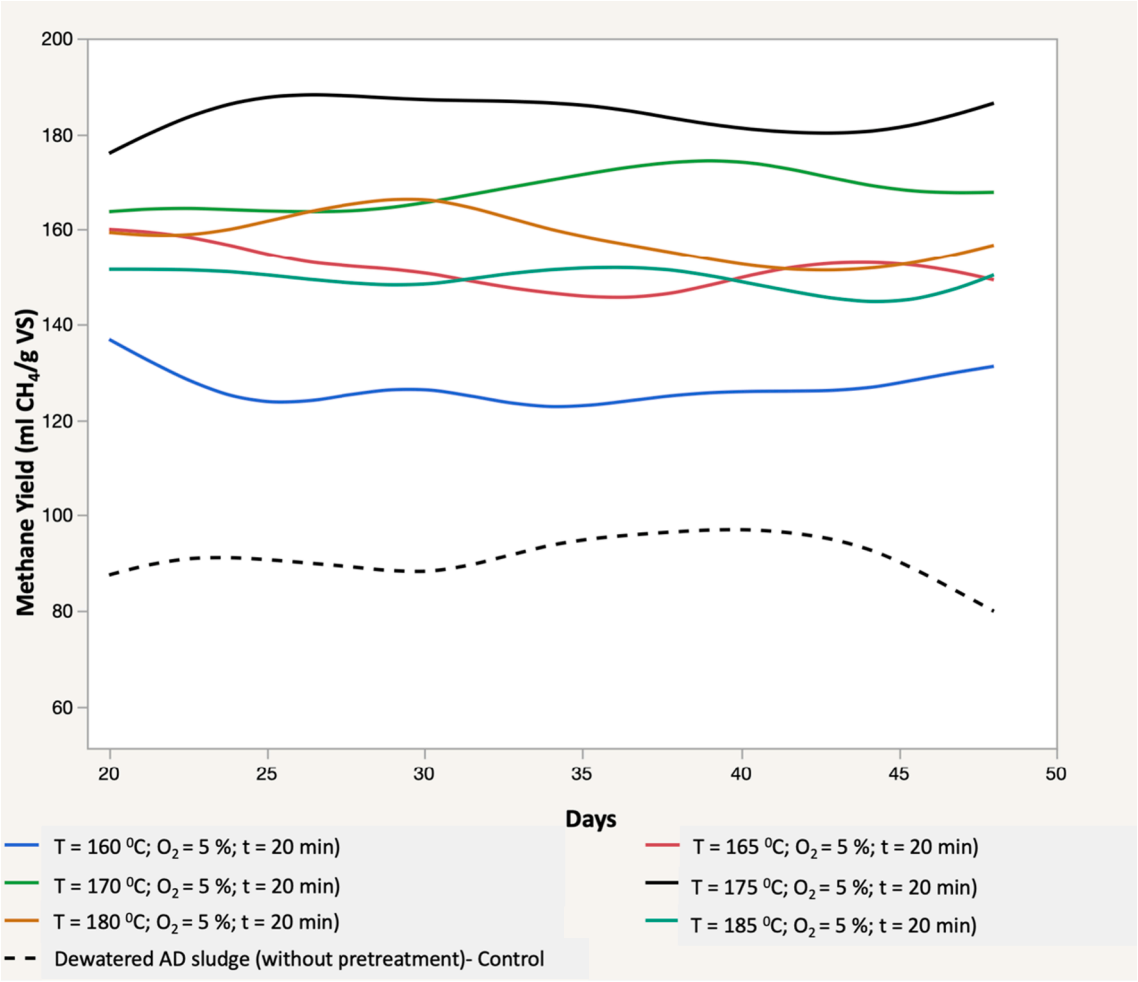


Fig. 2. Average methane yield of bioreactors fed with AWOEx pretreated DSS material pretreated with 5% oxygen.

were determined using an Universal gas analyzer, UGA series (Stanford Research Systems, Sunnyvale, CA) (Khan & Ahring, 2020). 10 mL of gas sample was collected daily before the reactors were fed and injected into the gas analyzer using a peristaltic pump.

3. Results and discussion

3.1. Effect of the AWOEx pretreatment on DSS

The characteristics of DSS are shown in Table 2. DSS contained 86.44 % of moisture contents (MS), 13.56 % of total solids (TS), and 5.78 % of volatile solids (VS) which is 77.8 % of TS. The results were similar to previously reported data (Nordell et al., 2021; Usman et al., 2019). Further, it was found that DSS contained 2.3 % cellulose, 2.9 % hemicellulose, 0.9 % lignin, 1.69 % of crude fat and 3.01 % of protein.

Based on the statistical analysis of the experimental data for both the tested categorical variables, addition of oxygen significantly improved methane yield for all the temperatures ranges of the pretreatment ($p < 0.0003$). On the other hand, temperature improved methane yield ($p < 0.057$)- but not to an extent as seen in case of oxygen addition.

We calculated the Cohen's D to find the implication of the effect size on our study. If two groups are the same size and have equal standard deviations, Cohen's d is the suitable effect size measurement. If each group has a distinct standard deviation, an alternative metric is Glass's delta, which only considers the standard deviation of the control group. An option when there are diverse sample sizes is Hedges' g, which offers a measure of effect size weighted in accordance with the relative size of each sample. In our study, Cohen's d was calculated to be 4.22; Glass's *delta* was calculated to be 4.41 and Hedges' *g* was 4.23.

3.1.1. Effect of the AWOEx pretreatment on methane yield from DSS

At the end of HRT 3, the average biogas aaition of methane and carbon dioxide were in the range of 62 ± 1.5 vol% and 38 ± 2.2 vol%ˆ, respectively, with traces of moisture, hydrogen and oxygen. For the control (DSS without pretreatment), methane and carbon dioxide content were in the range of 55–58 vol% and 40–42 % Fig. 1 shows the methane yield from AWOEx pretreated DSS material at temperatures ranging from 160 °C – 185 °C with 20 min residence time and without oxygen. Fig. 2 shows the trend in average methane yields from the semi-continuous bioreactors at 55 °C fed with DSS pretreated at temperatures ranging from 160 °C to 185 °C with 20 min residence time and 5 % oxygen. The control reactors were fed with DSS without AWOEx pretreatment to compare the effect of the different conditions during AWOEx pretreatment on methane production. It was found that the average methane yield of the control reactors was 92.92 mL/gVS. The average methane yields from 175 °C and 170 °C were 183.91 and 170.93 mL/g vS respectively. Since the residence time and oxygen dosage were kept constant for the optimally performing sets, it can be concluded that the resulting methane yield was solely a function of pretreatment temperature.

The average methane yield at 175 °C was higher than 180 °C, 170 °C and 165 °C by 18.72 %, 7.59 %, and 23.77 %, respectively. One of the possible reasons of lower methane yield at 180 °C might be due to the formation of inhibitory compounds at higher temperatures as previously

Table 3
TS and vS % of the thermophilic bioreactors at different AWOEx pretreatment condition after 1,2 and 3 retention times (HRTs).

Run	Conditions	TS (%) [in]	TS (%) [out] [HRT 1]	TS (%) [out] [HRT 2]	TS (%) [out] [HRT 3]	vS (%) [in]	vS (%) [out] [HRT 1]	vS (%) [out] [HRT 2]	vS (%) [out] [HRT 3]
1	T = 160 °C; t = 20 min; O_2 = 0 %	9.17	8.22	6.56	6.55	74.33	70.77	69.49	67.12
2	T = 165 °C; t = 20 min; O_2 = 0 %	9.12	8.25	7.44	7.22	74.66	72.21	71.84	69.45
3	T = 170 °C; t = 20 min; O_2 = 0 %	9.08	8.87	7.57	7.15	76.89	75.19	74.91	71.35
4	T = 175 °C; t = 20 min; O_2 = 0 %	9.17	7.03	7.12	6.97	74.26	71.78	70.73	69.92
5	T = 180 °C; t = 20 min; O_2 = 0 %	9.07	6.93	5.22	4.97	73.69	72.29	70.71	71.34
6	T = 185 °C; t = 20 min; O_2 = 0 %	9.32	5.89	4.49	4.33	74.26	73.91	73.95	72.90
7	T = 160 °C; t = 20 min; O_2 = 5 %	9.22	7.34	5.30	3.26	73.80	70.45	69.55	68.92
8	T = 165 °C; t = 20 min; O_2 = 5 %	9.17	7.17	5.49	5.01	78.57	75.88	74.34	72.92
9	T = 170 °C; t = 20 min; O_2 = 5 %	9.24	8.6	5.12	4.72	73.04	72.35	73.92	70.23
10	T = 175 °C; t = 20 min; O_2 = 5 %	9.12	5.54	4.01	2.48	74.14	72.62	66.41	65.72

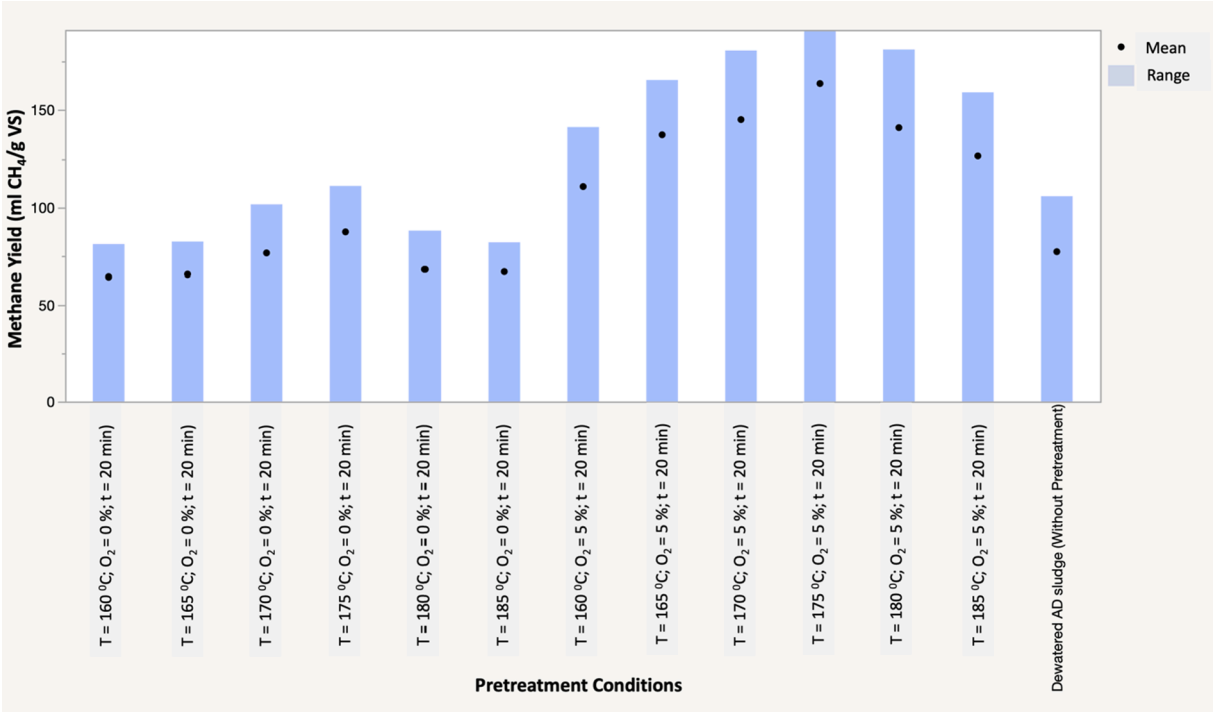


Fig. 3. Comparing the effect of the two tested pretreatment variables on methane yield: DSS pretreated with and without oxygen at temperatures ranging from 160 to 185 °C. Range of the methane yield is plotted with mean marked for each pretreatment condition (bar).

described during the pretreatment of sewage sludge (Hii et al., 2014; Suárez-Iglesias et al., 2017). The highest average methane yields of 183.92 mL/gvS at 175 °C and 170.93 mL/gvS at 170 °C was for the same residence time and O_2 dosage but at different temperatures, indicating that lower temperatures might have resulted in a lower hydrolysis of the material than higher temperatures. The results for the control reactor are in accordance with the study by Nordell et al., 2021, which found that only 46 mL/g vS of methane yield was achieved during anaerobic digestion of digested sewage sludge under thermophilic condition without any thermochemical pretreatment. Overall, AWOEx pretreatment of DSS significantly increased the methane yield far over the control reactor based on the ANOVA test (P = 0.0007). Table 3 shows the difference of TS and vS % of all bioreactors for the feed and the effluent, and it was found that the bioreactors with the highest methane yield, 175 °C, had the highest TS conversion of 72.82 % out of all the groups. Highest single day average methane yield was 191.4 mL/g vS and 181.3 mL/g vS observed at the pretreatment temperature of 175 °C and 180 °C, respectively, followed by 180.8 mL/g vS at 170 °C.

Based on our results, optimal temperature range for pretreating DSS will be in the range of 170–180 °C with oxygen addition. Fig. 3 shows the mean methane yield plotted for every pretreatment condition where oxygen addition resulted in an increase of the methane yield by 92.65 % over the pretreatment groups without oxygen addition at 175 °C, the temperature with the maximum average methane yield. The increasing trend of methane yield with an increase in temperature for 160 to 175 °C and decreasing trend from 175 to 185 °C, further supports that 175 °C is the optimal temperature for pretreating DSS.

3.2. *Influence of the different operational conditions of AWOEx pretreatment of DSS on methane yield*

Oxygen addition had significant (P = 0.0003) and higher effect on methane yield next to the temperature (P = 0.0573) as seen in Fig. 3. This is in accordance with a previous study on AWOEX pretreatment of sawdust (Biswas et al., 2020; Higgins et al., 2017). A previous study reported that AWOEx pretreatment increased the formation of small molecular weight organics, which is easily biodegradable during AD (Hii et al., 2014; Suárez-Iglesias et al., 2017). Further, the radicals produced during AWOEx pretreatment are powerful in oxidizing recalcitrant compounds such as lignin, which increased the biodegradability of DSS

Table 4
Characteristics of influents and effluents of the thermophilic semi-continuous bioreactors.

Sr. No	Thermophilic Reactor (Inlet material)	Protein (%) in	Protein (%) out	Lignin (%) in	Lignin (%) out	Cellulose (%) in	Cellulose (%) out	Hemicellulose (%) in	Hemicellulose (%) out	Crude fat (%) in	Crude fat (%) out
1	T = 160 °C; t = 20 min; O_2 = 0 %	2.81	2.29	1.15	0.6	4.91	3.54	3.52	1.61	0.73	0.35
2	T = 165 °C; t = 20 min; O_2 = 0 %	2.89	2.53	1.03	0.47	5.15	3.51	3.67	1.73	0.71	0.47
3	T = 170 °C; t = 20 min; O_2 = 0 %	2.22	1.99	0.98	0.45	5.28	3.61	3.85	1.78	0.72	0.33
4	T = 175 °C; t = 20 min; O_2 = 0 %	2.17	1.86	0.92	0.55	5.45	3.52	3.98	1.8	0.70	0.40
5	T = 180 °C; t = 20 min; O_2 = 0 %	2.02	1.76	0.88	0.51	5.7	3.44	4.01	1.89	0.75	0.44
6	T = 185 °C; t = 20 min; O_2 = 0 %	2.91	2.81	0.90	0.52	5.85	3.39	4.16	1.87	0.73	0.31
7	T = 160 °C; t = 20 min; O_2 = 5 %	2.72	2.51	0.88	0.51	4.65	1.8	2.99	1.5	0.77	0.28
8	T = 165 °C; t = 20 min; O_2 = 5 %	2.84	2.31	0.82	0.46	4.78	1.79	3.15	1.52	0.73	0.25
9	T = 170 °C; t = 20 min; O_2 = 5 %	1.75	1.51	0.80	0.34	4.93	1.51	3.20	1.31	0.73	0.19
10	**T = 175 °C; t = 20 min; O_2 = 5 %**	**1.68**	**1.39**	**0.76**	**0.29**	**4.99**	**1.28**	**3.46**	**1.15**	**0.75**	**0.16**
11	T = 180 °C; t = 20 min; O_2 = 5 %	1.86	1.34	0.74	0.41	5.10	1.67	3.50	1.78	0.76	0.21
12	T = 185 °C; t = 20 min; O_2 = 5 %	2.78	2.22	0.70	0.36	5.35	2.24	3.67	2.18	0.71	0.19
13	Control (Dewatered AD Sludge without Pretreatment)	3.01	2.67	1.76	1.47	4.45	3.54	3.16	1.69	1.3	0.83

(Khan & Ahring, 2020). Anaerobic digestion breaks down long-chain molecules into shorter chains (Abelleira-Pereira et al., 2015; Hii et al., 2014; Liang et al., 2020; Lin et al., 2022; M'Arimi et al., 2020; Suárez-Iglesias et al., 2017). Furthermore, higher O_2 pressure during pretreatment rapidly depolymerizes crosslinked bonds of complex compounds into simpler chain. The breakdown of bonds is somewhat constrained in the absence of oxygen. According to our findings, high temperatures enhance the dissolution of DSS, and adding O_2 helps in oxidizing organic compounds in DSS into digestible molecules.

3.3. Composition analysis of DSS before and after AWOEx pretreatment-AD

AWOEx treatment affected the composition of DSS significantly. At the optimal pretreatment temperature of 175 °C, the conversion of cellulose was 74.4 %, hemicellulose was 66.8 %, and lignin was 68.8 % of the material. However, for the control set (dewatered DSS) only 28.6 % of the cellulose, 51.2 % of the hemicellulose and 47.2 % of the lignin was converted during secondary thermophilic AD. Table 4 shows the composition analysis of the influent and effluent of the bioreactors between 160 and 185 °C compared to the control bioreactor.

The average protein content decreased with 17.3 % and 28.1 % in the bioreactors at 175 °C and 180 °C, respectively, compared to 11.3 % protein conversion in the control reactor Thermal hydrolysis do not only lead to conversion of lipids, proteins and polysaccharides but simultaneously brings about hydrolysis of these compounds into lower molecular weight compounds by cleaving the peptide, ester and glycosidic bonds. The crude fat content in the effluent of our bioreactors fed with material pretreated at175°C and 180 °C decreased by 78.66 % and 72.36 %, respectively, compared to 36.15 % reduction in the control bioreactor set. In the effluent sets of the bioreactors 175 °C and 180 °C without oxygen, the reduction in crude fat content was 42.85 % and 41.33 % showing that oxygen triggers an aggressive breakdown of crude fats in DSS. This result indicate that highly unsaturated fatty acids are more susceptible to oxidative reactions. Polymers in the material need to be hydrolyzed in order to allow for methane production by anaerobic digestion. Grease and fat hydrolysis into long chain fatty acids, which can be further degraded by AD possess a large biogas potential of importance for the methane yield. (Wett et al., 2006).

A recent study reported a hydrolysis of ca. 30 % of the total proteins and 25 % of the total polysaccharide content into simpler, smaller organics following thermic pretreatment (Yang et al., 2019). When thermal hydrolysis temperatures were increased from 120 to 170 °C, (Chen et al., 2019) reported a 15 % increase in amino acid hydrolysis as compared to a 9.7 % increase in proteins hydrolyzed to VFA. Another study reported a 60 % increase in the amount of VFA produced from protein hydrolysis when the temperature was increased from 160 to 200 °C(Wilson & Novak, 2009). Another study reported that during thermal hydrolysis at 160 °C for 50 min, around 12 % of particulate protein was transformed into VFA (Han et al., 2017). It has been estimated that 13 to 23 % of the total VFA generated in wastewater sludge may be generated from the hydrolysis of protein (Wilson & Novak, 2009).

Studies also indicate that lipid hydrolysis at ca. 170 °C primarily contained fractions of aldehydes(80 %) and alcohol(11 %) (Wilson & Novak, 2009). During thermal hydrolysis, long-chain fatty acids (LCFA) increase in concentration, inhibiting bacteria and methanogens, although the inhibitory concentration depends on the LCFA composition and microbial species composition. In contrast to saturated fatty acids, unsaturated LCFAs were more inhibitive to methanogens at threshold concentrations of 10 to 30 mg/L (Lalman & Bagley, 2000).

The presence of lipids and proteins in wastewater sludge may generate compounds which are inhibitory to methanogenesis such as ammonia (Liu & Sung, 2002; Tanaka et al., 1997) and hydrophobic fatty acids (Angelidaki & Ahring, 1992; Chowdhury et al., 2007) through the thermal hydrolysis of proteins and lipids, respectively. Under the optimal conditions leading to the highest methane yield, the biodegradation of the organic content of the DSS significantly improved. A

Table 5
VFAs analysis of influents and effluents of the thermophilic semi-continuous bioreactors.

VFAs (mg/L)	Inlet stream			
	175 °C; 20 min; 5 %	180 °C; 15 min; 5 %	175 °C; 20 min; 0 %	180 °C; 15 min; 0 %
Acetic acid	900	860	675	830
Propionic acid	240	330	220	210
Isobutyric acid	190	125	175	110
Butyric acid	200	150	100	120
Isovaleric acid	120	95	90	100
VFAs (mg/L)	**Outlet stream**			
Acetic acid	15	8.0	10.5	8.0
Propionic acid	2.0	1.25	2.50	1.50
Isobutyric acid	1.40	0.95	1.30	1.40
Butyric acid	1.50	1.30	1.20	1.52
Isovaleric acid	1.00	1.80	1.30	1.10

previous study also reported that AWOEX pretreatment showed enhanced solubilization of carbohydrates after AWOEx pretreatment, which resulted in a higher methane yield during AD of pretreated manure fibers (Khan & Ahring, 2020). In this study, we further saw improved conversion of other components such as proteins and lipids. As a result of the hydrolysis reaction of acetyl groups in hemicellulose producing acetic acid, the pretreated DSS after AWOEx at 175 °C contained higher concentrations of VFAs (1650 mg/L) (Jönsson & Martín, 2016). Acetic acid dominated the organic acids in all bioreactors, and the bioreactor that contained the highest concentration of VFA also contained the highest concentration of acetic acid (900 mg/L) (Table 5). The concentration of VFA in the effluents of all bioreactors were, however, low (<20 mg/L), indicating that all bioreactors are functioning well and all digestible materials are being fully converted to methane.

One of the major findings of our study was the degradation of crude fat and protein in the optimally performing set of reactors at 175 °C by ca.80 % and 20 %, respectively. It is obvious that this increased conversion is a major reason for the higher methane yield found after AWOEx. Overall, our results show that AWOEx pretreatment of DSS results in significant higher methane production compared to standard one-step AD of sewage sludge. This higher energy recovery is of major importance for valorizing sewage sludge at WWTPs. The study further demonstrated that both elevated temperature and oxygen are important for improving the carbon conversion of sewage sludge into methane. Compared to conventional AD with ca. 50 % vS conversion, incorporating a secondary AWOEx pretreatment process resulted in 41 % extra conversion of organics in sewage sludge with over 100 % net improved methane production. This far higher methane production as well as the lower residual amount residing needing final disposal could potentially be game-changing for exploitation of the energy potential of sewage sludge in the future.

4. Conclusions

This study demonstrates the efficiency of AWOEx for enhancing methane yield of AD of DSS demonstrating a 2-fold increase in methane yield compared to untreated DSS far higher than previously reported in other studies. The optimal pretreatment condition was found to be a temperature of 175 °C with oxygen (5 % of VS) added resulting in a methane yield of 183.91 ± 4.93 mL/g vS of previous digested sewage sludge. Oxygen addition was the most critical parameter in improving the methane yield, followed by temperature. Pretreatment with a secondary AD step in tandem improved the CCE significantly leaving only small amounts of organics (15 % of the input VS) behind. From this study we also demonstrated that AWOEx not only had a major effect on the digestibility of lignocellulosic polymers such as carbohydrates, which was present in low amounts in the DSS but also on the lipids and protein fraction, which was present in large concentrations in the sewage sludge.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was funded by the DOE BETO under the award DOE EERE BETO DE-EE0008933. The authors thank Muhammad Usman for technical assistance and the PNNL-WSU Distinguished Graduate Research Program for support to Anthony Giduthuri.

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